

Traffic Noise Monitor — Quick-Start Guide

Turnkey Raspberry Pi Setup with Docker & Web Dashboard

Version 1.2.0

What You Need Before Starting

- ✓ **Raspberry Pi** (3, 4, 5, or Zero 2W)
- ✓ **USB Microphone** (or soundcard)
- ✓ **Ethernet Cable** (for 2-min setup)

Step 1: Hardware Connections & 1-Step Installation

Terminal

1. Plug your **USB Microphone** and **Ethernet cable** into the Pi, then connect power.
2. Copy `noise-bot-friend.zip` to your Pi and run these commands in the terminal:

```
unzip noise-bot-friend.zip -d noise-bot-friend
cd noise-bot-friend
chmod +x install.sh
./install.sh
```

The installer configures Docker, maps microphone hardware (`/dev/snd`), and starts **Watchtower** for background auto-updates.

Step 2: Open the Web Dashboard

Phone / PC Browser

From any phone or laptop on your Wi-Fi, open:

`http://<your-pi-ip>:5000` or `http://noise-bot.local:5000`

Step 3: Web Configuration & Calibration

Settings Menu (Passcode: admin123)

1. Elevation & Distance Triangulation (Audio Tab):

- **Floor Number:** Select your floor (1 = Ground, 3 = 3rd Floor, 10 = 10th Floor).
- **Setback:** Enter horizontal distance (meters) from your building to the road.
- The app auto-triangulates the 3D line-of-sight sound ray and calculates exact acoustic loss to the vehicle tailpipe:

3rd Floor (6m height) + 5m setback → Direct Sound Ray: 7.8m (Loss: +23.9 dB)
10th Floor (27m height) + 10m setback → Direct Sound Ray: 28.8m (Loss: +35.2 dB)

2. Join Wi-Fi Network (Wi-Fi Tab):

- Tap **Scan Networks**, pick your Wi-Fi, type your password, and click **Join Wi-Fi**.
- Once connected, you can **unplug the Ethernet cable** and place the Pi wherever you like!

3. Bluesky Automated Posting (Alerts Tab — Optional):

- Toggle **Enable Bluesky**.
- In Bluesky (web/app): Settings → Privacy & Security → App Passwords → Add App Password .
- Paste your Handle (e.g. `mybot.bsky.social`) and generated App Password, then click **Save & Apply**.

Step 4: Monitoring, Stats & Re-Classification

Live Features

- **Live View:** Real-time sound pressure meter showing sensor level and estimated tailpipe level.
- **Stats View: Daily Events Bar Chart** showing noise violations per day over the past 14 days.
- **Audio Playback & Tagging:** Listen to 8-second clips and re-classify tags on the fly ( Traffic,  Siren,  Construction,  Weather,  Misc) using the dropdown on each clip.

 **Zero-Maintenance Auto Updates (Watchtower):** Checks for updates in the background every 5 minutes. Whenever new code is published, the bot auto-updates without erasing your saved settings or recordings!